

APS110 TUT2: Elastic Behaviour

SEPTEMBER 19, 2023
MICHELLE LIN | PRA103 & PRA 125

1 Announcements

- First graded quiz is on **Friday, Sept 22**
- Reminder that tutorial quizzes are not for credit

2 Content Review

- Definitions:
 - Young's modulus - E (MPa)
 - Engineering stress - σ (MPa)
 - Engineering strain - ϵ (unitless)
- Calculate any of them given the other two (take extra care with units!)

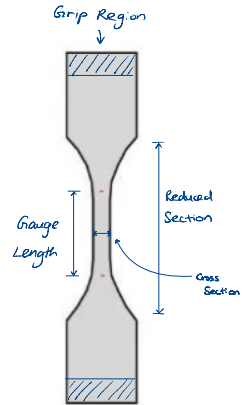
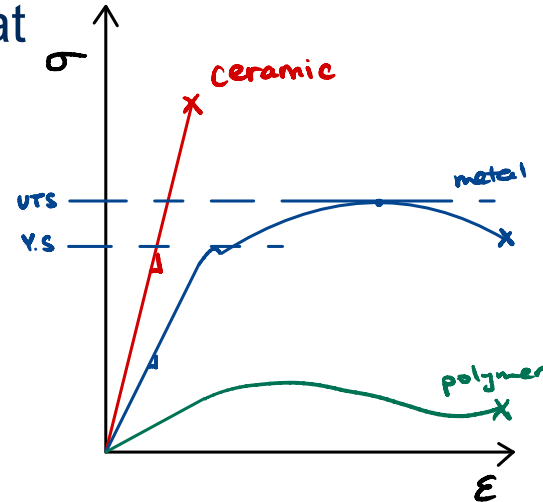
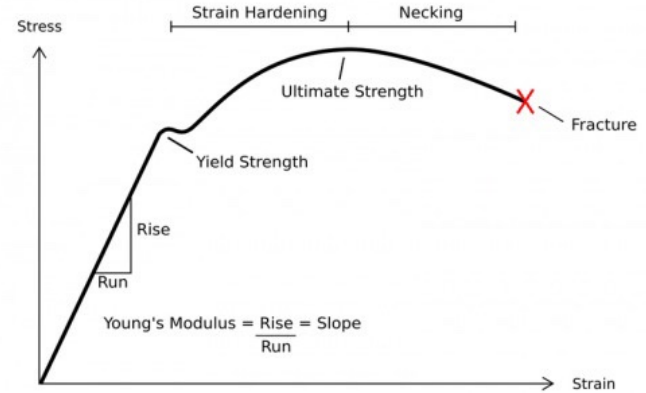
$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$$

$$\sigma = \frac{F}{A} ; \epsilon = \frac{\Delta l}{l}$$

Giga-	Mega-	Kilo-
10^9	10^6	10^3

2 Content Review

- Stress-strain curve:
 - General appearance
 - Generated from a tensile test, how tensile tests are performed, curve is the elastic region, what are the
 - Parts of a standard tensile specimen



3 Practice Quiz (1/6 Questions) - 1 min

1. Among the material classes listed below, which would you expect to have the highest Young's modulus? (Can you sketch a generalized stress-strain curve for each class?)

- (a) Ceramics
- (b) Metals
- (c) Polymers

3 Practice Quiz (2/6 Questions) - 30 sec

2. Which of the following are the correct units of strain?

- (a) m
- (b) mm
- (c) inches
- (d) Strain is dimensionless

Can be expressed as decimal or percentage

3 Practice Quiz (3/6 Questions) - 30 sec

3. Why, in the characterization of materials, engineering strain and engineering stress are used rather than force and elongation?
- (a) Because they are easier to calculate.
 - ☒ (b) Because they are sample size independent.
 - (c) Because they depend on sample size, such as length and initial area.
 - (d) None of the above.

3 Practice Quiz (4/6 Questions) - 2 min

4. A cylindrical rod with a diameter of 12.5 mm and a length of 600 mm is loaded in tension. The rod is expected to not elongate more than 1.4 mm under an applied load of 29 kN. If the Young's modulus of Aluminum is 70 GPa, can this rod be made of Aluminum? What would be the elongation expected for it in that case?
- (a) Yes, because the elongation would be of only 1.01 mm.
 - ☒ (b) No, because the elongation would be of 2.02 mm.
 - (c) No, because the elongation would be of 20.2 mm.
 - (d) No, because the elongation would be of 0.00337.

$$E = \frac{\sigma}{\epsilon} = \frac{F/A}{\Delta l/l} \Rightarrow \Delta l = \frac{F \times l}{A \times E} = \frac{(29 \times 10^3 \text{ N})(600 \times 10^{-3} \text{ m})}{\pi \frac{(12.5 \times 10^{-3} \text{ m})^2}{4} \times 70 \times 10^9}$$

$$\Delta l = 2.02 \text{ mm}$$

3 Practice Quiz (5/6 Questions) - 30 sec

5. Which of the following can be considered a scientific model?
- (a) Atoms as hard spheres
 - (b) Net interatomic force as a mechanical spring
 - ☒ (c) Both *a* and *b*
 - (d) None of the above

3 Practice Quiz (6/6 Questions) - 2 min

6. Consider a person attempting to pop a balloon with a series of nails on a wooden board, assuming they are only able to apply a maximum force of 100 N with their arms. A balloon typically bursts when a pressure of 110 kPa is applied to its surface. Considering that the person could effortlessly pop the balloon with a single nail when applying 5.5 N, what is the minimum number of nails for the person to be incapable of bursting the balloon? Consider that the pressure is equally distributed among the nails. (Hint: you can start by calculating the area of the tip of the nail. Alternatively, you can skip this calculation and solve the problem based on proportionality.)

- (a) 17
- (b) 18
- ☒ (c) 19
- (d) 20

$$\frac{5.5 \text{ N}}{1 \text{ nail}} = \frac{100 \text{ N}}{\text{nail max}}$$

$$\text{nail max} = \frac{100 \text{ N}}{5.5 \text{ N}}$$

$$\text{nail max} = 18.18$$

4 Sample Problems (1/2 Questions) - 10 min

1. A cylindrical rod with a diameter of 12.5 mm and a length of 600 mm is loaded in tension. It is made of brass, which has a Young's modulus of 100 GPa. The force applied to it is 2 kN. *Soln in other PDF*
 - (a) What is the transversal area upon which the force is being applied?
 - (b) What is the stress on the rod?
 - (c) What is the strain experienced by the rod? What is its length while this force is being applied?
 - (d) Consider a second rod with the same length and a diameter of 25.0 mm, also made of brass. What is the Young's modulus of this rod?
 - (e) This second rod is loaded in tension. If we want it to experience the same strain as the first rod, what should be the applied force?

4 Sample Problems (2/2 Questions) - 5 min

2. Sketch a figure for the net interatomic forces versus interatomic spacing between two atoms (r). Clearly label the axes and indicate the shape of the curve for the attractive forces F_a , repulsive forces F_R , and net force F_{net} .

Soln in PDF.

- (a) Indicate the source of each of these forces and discuss the significance of the point where the net force curve intersects the horizontal axis ($r = r_0$).
- (b) Explain the relationship of Young's modulus to this curve and why it is a structure-independent property.